MATHEMATICS
Introduction to Modern Algebra I
Quiz 5.

1) Let G be a group. For any $a \in G$ define $C(a) = \{g \in G | ga = ag\}$. Show that $C(a)$ is a subgroup of G.

Proof. We must check that $C(a)$ is non-empty, closed under multiplication and closed under inverses.

$eg = ge$ and so $e \in C(a)$. In particular $C(a) \neq \emptyset$.

If $g, h \in C(a)$, then $ga = ag$ and $ha = ah$. Thus $(gh)a = g(ha) = g(ah) = (ga)h = (ag)h = a(gh)$. Hence $C(a)$ is closed under multiplication.

If $g \in C(a)$, then $ga = ag$ thus $g^{-1}a = g^{-1}agg^{-1} = g^{-1}gag^{-1} = ag^{-1}$ and hence $g^{-1} \in C(a)$. Hence $C(a)$ is closed under inverses and so it is a subgroup of G. $\square$

2) Let H be a subgroup of a group G. If $a, b \in G$, define $a \sim b$ if $ab^{-1} \in H$. Show that $\sim$ is an equivalence relation.

Proof. Reflexive: $aa^{-1} = e \in H$ because subgroups always contain e.

Symmetric: If $a \sim b$ then $ab^{-1} \in H$. But then $ba^{-1} = (b^{-1})^{-1}a^{-1} = (ab^{-1})^{-1} \in H$. Thus $b \sim a$.

Transitive: If $a \sim b$ and $b \sim c$, then $ab^{-1} \in H$ and $bc^{-1} \in H$. But then $ab = aeb = ab^{-1}bc^{-1} \in H$. $\square$